Supplementary Methods

Detecting outliers of pursuit eye movements:

a preliminary analysis of autism spectrum disorder.

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Part1: Time series analysis of pursuit in slow Lissajous task

1.1. Coordinate System for Lissajous Trajectory

Since the visual target moved along a complex Lissajous curve, we analyzed eye movements using a polar coordinate system relative to the center of the display (Supplementary Figure S1). This approach is more robust than Cartesian coordinates for evaluating pursuit performance on curvilinear paths. For each data point (sampled at 1 ms), the radial distance from the center was calculated for both the target and the participant's gaze.

1.2. Temporal Deviation (Δt)

To quantify the timing error, we calculated the temporal lag (Δt) at each data point (Supplementary Figure S2). This was defined as the time difference required for the participant's gaze to reach the same radial position as the target.

Calculation: For each data point, at time t, performed a subtraction between the time the target occupied a specific radial coordinate, and the time the participant's gaze reached that same coordinate.

Significance: A positive Δt indicates a pursuit lag (delay), while fluctuations in Δt reflect poor predictive control and temporal instability.

1.3. Spatial Deviation (Δs)

Spatial deviation was quantified as the difference between the radial distance of the target and that of the gaze from the center of the monitor at each 1 ms interval (Supplementary Figure S3).

Calculation: This spatial shift (Δs) was calculated by subtracting the target's radial distance from the participant's gaze radial distance and subsequently normalized by the instantaneous radial distance of the target.

Significance: A larger Δs reflects a mismatch in amplitude control, indicating a significant spatial gain error of the pursuit movement relative to the target's trajectory.

1.4. Integration into the Outlier Score

The final "Outlier Score" was derived by integrating these Δt and Δs values across the entire trial duration. By aggregating these point-by-point subtractions into a single composite metric, we were able to capture subtle idiosyncratic patterns—such as individuals who maintain spatial accuracy but with a significant temporal lag, or vice versa.

This individualized quantification allows for the detection of ASD-specific "atypicality" that is often masked in group-mean analyses.

Part2: Outlier analysis

2.1. Rationale for Outlier Analysis

Conventional group-level comparisons, such as t-tests, focus on differences in mean values. However, these methods may not adequately capture the clinical heterogeneity inherent in autism spectrum disorder (ASD). In contrast, outlier analysis identifies individuals whose neurocognitive profiles deviate significantly from a normative distribution. This approach allows for the quantification of "atypicality" at the individual level, providing a more sensitive metric for characterizing the diverse oculomotor features associated with ASD.

2.2. Variable Selection and Dimensionality Reduction

To optimize the sensitivity of our outlier analysis, we initially considered four spatial parameters. We performed a Principal Component Analysis (PCA) to achieve dimensionality reduction and identify the most informative features for characterizing deviations. During this process, we evaluated the contribution of each variable to the principal components. We found that the mean value of Δs had negligible contribution rates. Consequently, we excluded the low contribution mean Δs and focused on other

parameters for the final outlier score calculation. This approach ensures that our composite metric prioritizes the most idiosyncratic aspects of oculomotor deviation.

2.3. Calculation of the Outlier Score

To derive a comprehensive measure of deviation, we calculated an "outlier score" for each participant based on smooth pursuit eye movement (SPEM) features. The procedure was as follows:

Calculation of features: We calculated temporal and spatial parameters, resulted in three basic parameters.

Normalization: For each selected index (x_i), we transformed the raw values into standardized Z-scores based on the mean (μ_{TD}) and standard deviation (σ_{TD}) of the typically developed (TD) group:

$$Z_i = \frac{x_i - \mu_{TD}}{\sigma_{TD}}$$

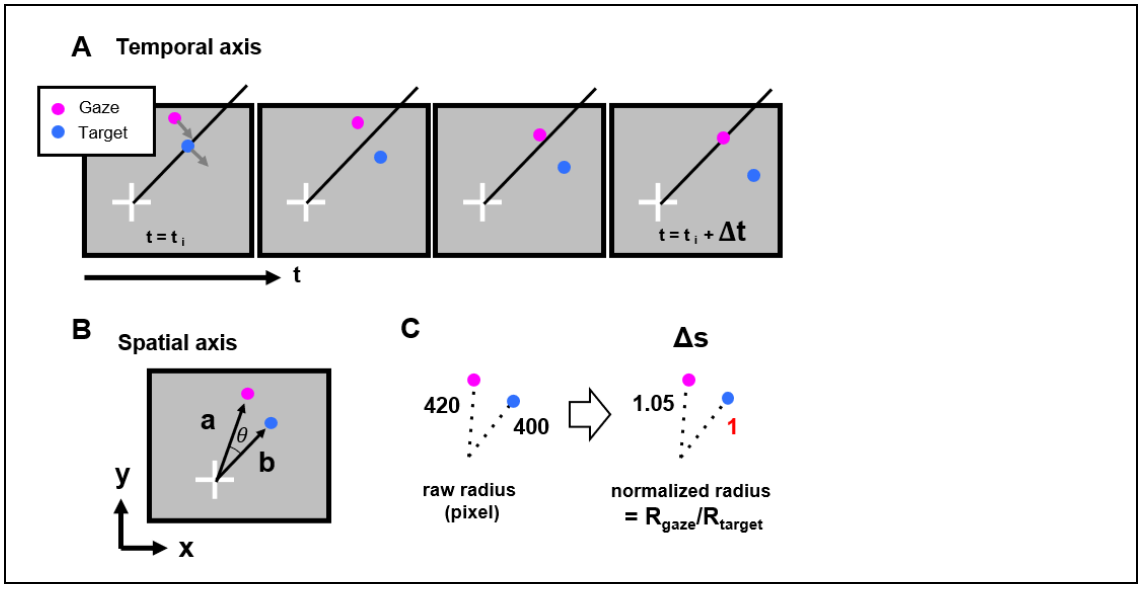
Aggregation: To evaluate the overall deviation across multiple dimensions, the individual Z-scores were aggregated by Principal Component Analysis and squared form of Mahalanobis distance. The resulting total value represents the "outlier score." A higher score indicates a more pronounced departure from the normative oculomotor patterns observed in the TD group.

2.4. Age-Dependency and Baseline Correlation

Given that oculomotor performance can be influenced by biological aging, we evaluated the relationship between age and outlier scores (Supplementary Figure S4). In the TD group, Pearson's correlation analysis with robust fit option was performed to establish a baseline for typical age-related changes. This allowed us to ensure that the elevated outlier scores in the ASD group were not merely a reflection of age-related decline but were specific to the clinical characteristics of the disorder.

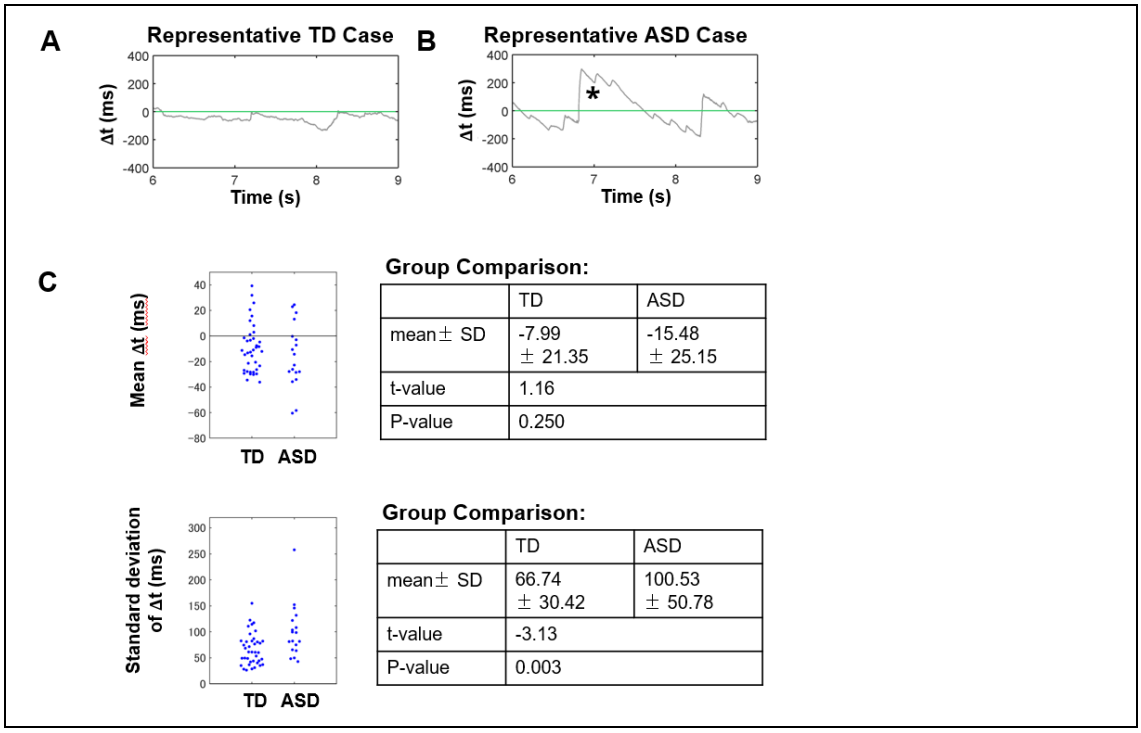
Statistical Robustness: Linear regression models were implemented using the robust fit option (fitlm with 'RobustOpts' in MATLAB). This method employs iteratively reweighted least squares to minimize the influence of potential outliers on the regression line, ensuring that our conclusion regarding the independence of age and outlier scores is statistically sound and not skewed by extreme values.

Supplementary Figures



Supplementary Figure S1

Geometric definitions for temporal and spatial deviation analysis. **(A)** Temporal Deviation (Δt): The timing error was quantified by defining a radial reference line passing through the target at time $\mathbf{t_i}$. The temporal deviation (Δt) represents the time required for the participant's gaze to cross this same reference line. A positive value indicates a pursuit lag. **(B)** Radial Distance Calculation: At each 1-ms interval, the radial distances from the monitor center to the target (**a**) and to the gaze (**b**) were calculated. **(C)** Spatial Deviation (Δs): To adjust for varying target eccentricity, the gaze radius was normalized by the target radius (Normalized Radius = R_{gaze}/R_{target}). The spatial deviation (Δs) reflects the gain error relative to this normalized baseline.



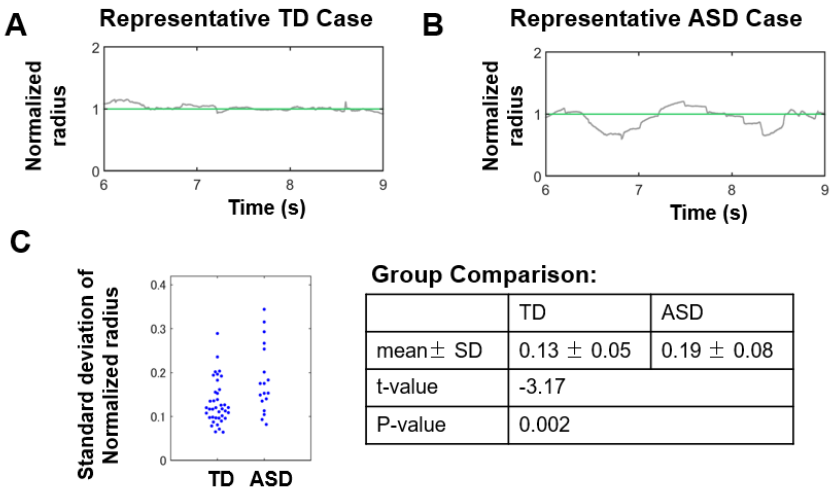
Supplementary Figure S2 Time-series analysis and group distribution of temporal deviation (Δt)

The temporal deviation reflects the stability of predictive timing during pursuit.

(A) Representative TD Case: The Δt remains stable and close to zero, indicating consistent predictive synchronization with the target trajectory. The raw gaze (gray) and target (green) traces are well-aligned in the time domain.

(B) Representative ASD Outlier Case: This participant exhibits marked temporal instability. The Δt fluctuates rapidly between lagging (positive values. *) and leading (negative values), representing a failure in stable predictive motor control.

(C) Distribution of Temporal Instability: Each dot represents the mean and standard deviation (SD) of Δt for an individual participant. The ASD group shows a wider distribution, with several individuals exhibiting SD values that fall significantly outside the range of the TD group.



Group Comparison:

	TD	ASD
mean± SD	0.13 ± 0.05	0.19 ± 0.08
t-value	-3.17	
P-value	0.002	

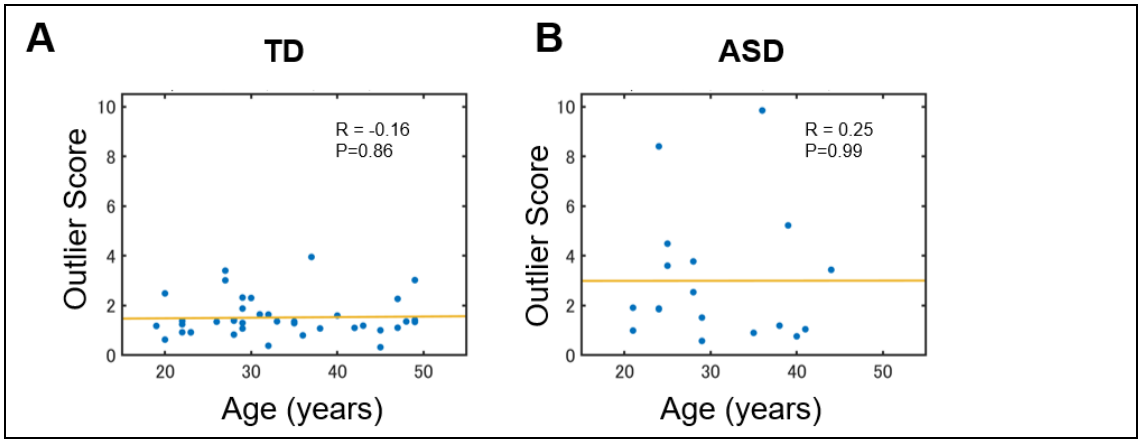
Supplementary Figure S3. Time-series analysis and group distribution of normalized spatial deviation (Δs)

The normalized spatial deviation reflects the accuracy of pursuit amplitude relative to the target's radial distance from the center. In **A** and **B**, The same participants as in Figure S2 are shown.

(A) Representative TD Case: The normalized radius remains stable and clustered near 1.0 (green)), indicating that the participant's gaze consistently matches the target's distance from the center. The raw gaze (gray) and target (green) traces overlap closely in the spatial domain.

(B) Representative ASD Outlier Case: This participant shows significant spatial instability. The normalized radius exhibits large, phasic fluctuations above and below 1.0, representing recurrent overshooting and undershooting of the target's radial position.

(C) Distribution of Spatial Instability: Each dot represents the standard deviation (SD) of the normalized Δs for an individual participant. The ASD group shows a higher concentration of individuals with elevated SD values, confirming a significant overrepresentation of spatial pursuit atypicality compared to the TD group.



Supplementary Figure S4. Relationship between age and outlier scores

To verify that the calculated outlier scores were not confounded by age, we performed correlation and regression analyses for both groups. **(A) Typically Developed (TD) Group:** No significant correlation was observed between age and outlier scores. The outlier scores remained stable across the adult age range, establishing a consistent normative baseline. **(B) Autism Spectrum Disorder (ASD) Group:** Similarly, age did not significantly influence outlier scores in the ASD group. High scores observed in specific individuals were not a product of age-related decline but rather reflected disorder-specific atypicality.